

Project README (Setup and Running)

This section provides the necessary instructions to set up the environment, install dependencies, and run the SmartLibrary application.

1. Prerequisites

You must have the following installed:

- **Python 3.x**
- **PostgreSQL Database Server**

2. Installation Steps

Step 2.1: Python Environment Setup

1. Create a Virtual Environment:

Bash

```
python -m venv smartlibrary_env
```

2. Activate the Environment:

Bash

On Windows (Command Prompt/PowerShell):

```
.\smartlibrary_env\Scripts\activate
```

On macOS/Linux:

```
source smartlibrary_env/bin/activate
```

3. **Install Python Dependencies:** Install the required dependencies: PyQt5, psycopg2-binary, and bcrypt.

Bash

```
pip install PyQt5==5.15.7 psycopg2-binary==2.9.6 bcrypt
```

Step 2.2: Database Setup

1. **Configure config.py:** Ensure the database connection details in your config.py file match your local PostgreSQL server settings.
2. **Create Database:** Manually create an empty database named smartlibrary in PostgreSQL.
3. **Run Migrations:** Execute the SQL statements from the **database_setup.sql** file against the configured smartlibrary database to create all required tables.

3. Running the Application

1. Ensure your virtual environment is **activated**.
2. Launch the application from the project root directory:

Bash

```
python gui.py
```

Test Credentials

The system manager seeds the database with initial accounts for testing:

Role	Email	Password
Librarian	admin@library.com	adminpass
Member	jane.member@uni.com	memberpass